ENVS 193DS Final

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Link GitHub repo

https://github.com/natty-nat/ENVS-193DS_spring-2026_final/tree/main

Set up

```
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(readxl) # reading .xlsx files
library(ggeffects)
library(DHARMa)
# storing nest data as object "nest"
nest <- read_csv(here("data", "nest_data_final.csv"))
personal_clean <- read_csv(
  here("data", "ENVS 193DS- Personal Data - Sheet1.csv"),
  # skipping the first 2 messed up rows
  skip = 2) |>
  # dropping the first empty column
  select(-1) |>
  # cleaning the names
  clean_names() |>
 # converting sleep to a graphable value
  mutate(
    sleep_hours =
      hour(hms(amount_of_sleep)) + (minute(hms(amount_of_sleep)) / 60),
    # converting the column "satisfaction" to numbers
    satisfaction = as.numeric(satisfaction))
```

Problem 1

1a.

In part 1 they used binary/ logistic regression.

In part 2 they used a Chi-Squared test for independence.

1b.

American Avocet microhabitat use is determined by the distance to the water's edge.

The shore birds American Avocet, Foster's Tern, and Black-necked Stilt, have preferences for certain habitats over others.

1c.

An additional variable that could effect American Avocet microhabitat choice could be their prey collected by looking in the soils. This is a continuous numeric variable (count/m^2), and it might influence habitat use because with limited prey the carrying capacity is lower (less food), and vice versa.

Problem 2

2a.

```r
# creating new object called "nests_clean" from "nest" data frame
nests_clean <- nest |>
  # selecting columns of interest
  select(case_control, height_cm, ant_species) |>
  # filtering to only include specific ant species
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # making new column for scientifc names and adjusting to names to be full
  mutate(scientific_name = case_when(
    ant_species == "AB" ~ "Crematogaster sjostedti",
    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"
  )) |>
  # reordering the levels of ant species
  mutate(scientific_name = factor(scientific_name, levels = c(
    "Crematogaster sjostedti",
    "Crematogaster mimosae",
    "Crematogaster nigriceps"
  )))
# displaying the structure of "nests_clean"
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ ant_species    : chr [1:270] "RRB" "RRB" "RRB" "RRB" ...
 $ scientific_name: Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
```

```
# showing 10 random rows!
nests_clean |>
  sample_n(10)
```

```
# A tibble: 10 x 4
   case_control height_cm ant_species scientific_name
          <dbl>     <dbl> <chr>       <fct>
 1            0      175  RRB         Crematogaster mimosae
 2            1      158. BBR         Crematogaster nigriceps
 3            0      200. RRB         Crematogaster mimosae
 4            0       93  BBR         Crematogaster nigriceps
 5            0      210. RRB         Crematogaster mimosae
 6            0      210  RRB         Crematogaster mimosae
 7            1      204. RRB         Crematogaster mimosae
 8            0      324. AB          Crematogaster sjostedti
 9            0      153  AB          Crematogaster sjostedti
10            0      199  RRB         Crematogaster mimosae
```

2b.

The 1 and 0 in the column "case_control" tell the reader whether or not the tree contained a nest. The 1s are affermative for nest, where the 0s show that there was no nest.

2c.

Ant species is a categorical variable with 3 levels. This could influence the probability of nest occurrence depending on the type of ant species, and how aggressive they might be to the nesting birds. I found this information in the introduction, paragraph 3.

Tree height is a continuous numerical value measured in cm. Tree height could influence the probability of nest occurrence because taller trees are more likely to have nests. I found this information in the methods section, paragraph 2.

2d.

Table 1: Table of models

Model Number	Model Type	Formula	Model Description
Model 1	Additive GLM	case_control ~ height_cm + scientific_name	Looking at the independent effect of tree height and ant species on nesting probability using logistic regression.
Model 2	Interactive GLM	case_control ~ height_cm * scientific_name	Looking at whether there is an effect of tree height on nesting probability changes with ant species present using logistic regression.

2e.

```
#running model 1: additive logistic regression
model_additive <- glm(case_control ~ height_cm + scientific_name,
                      data = nests_clean,
                      family = binomial)
```

2f.

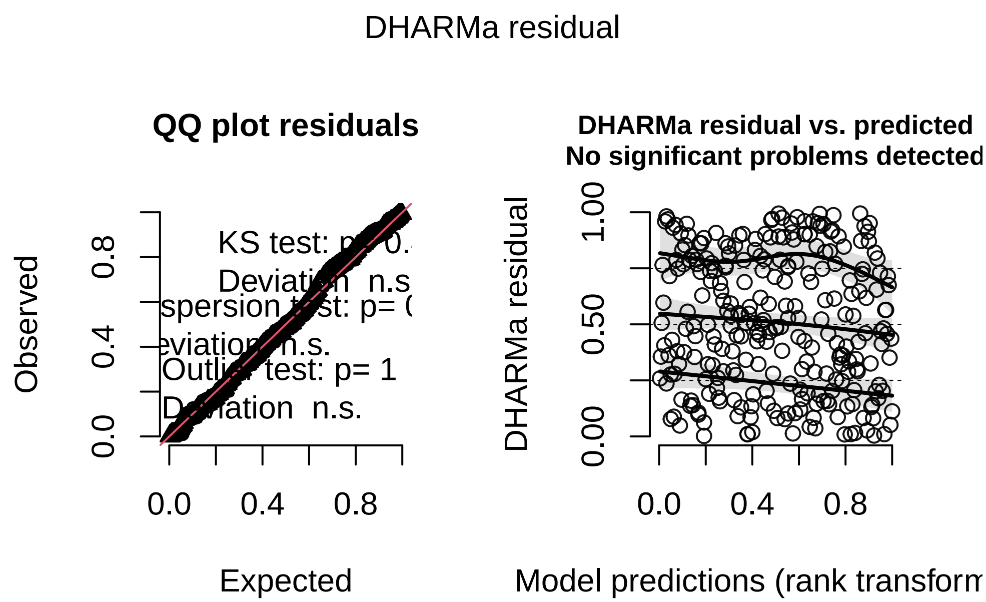
```
# Running IAC to check scores
AIC (model_additive, model_intereactive)
```

```
                   df      AIC
model_additive      4 258.7430
model_intereactive  6 237.8042
```

The best model as determined by Akaike's Information Criterion (AIC) is the interactive logistic regression model. This model has a lower AIC (237.8 compared to 258.7), which tells us there are significant changes to nesting depending on which ant species lives there in relation to tree height.

2g.

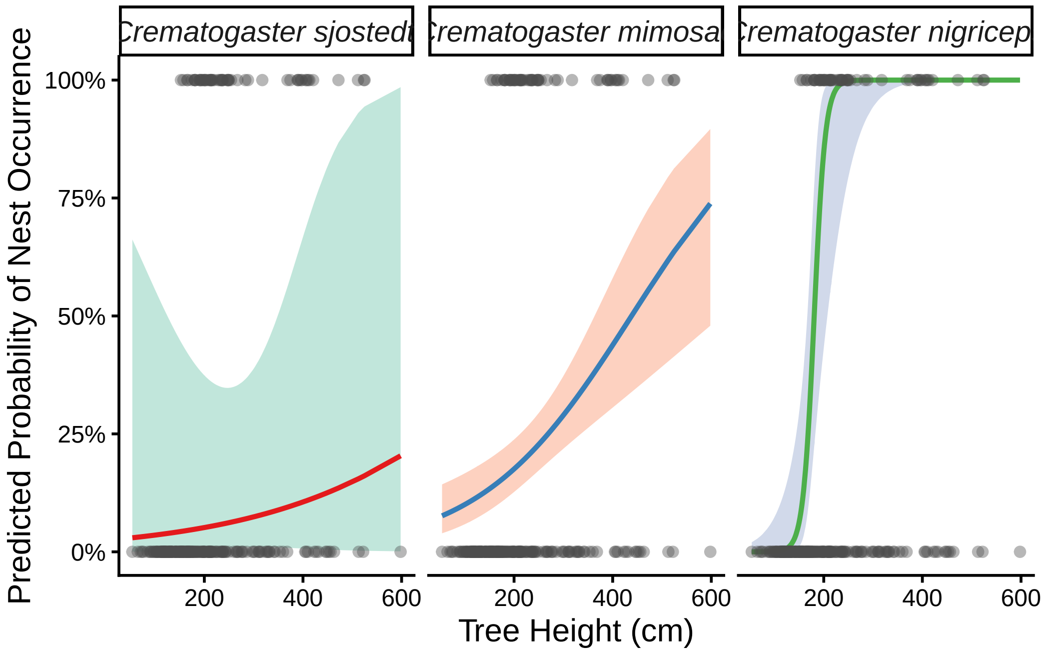
```
# simulate residuals for interactive model
sim_residuals <- simulateResiduals(fittedModel = model_intereactive, n = 250)
# plotting diagnostics
plot(sim_residuals)
```



2h.

```
# creating new object with predictions from interactive model
mod_preds <- ggpredict(model_intereactive,
                       terms = c("height_cm [all]", "scientific_name"))
# base layer: ggplot with predictions data
ggplot(mod_preds, aes( x = x,
                       y = predicted))+
  # first layer: geom ribbon to show confidence intervals for each group
  geom_ribbon(aes (ymin = conf.low,
                  ymax = conf.high,
                  fill = group),
              alpha = 0.4) +
```

```r
  # second layer: geom line for the model predictions
  geom_line(aes(color = group),
              size = 0.9)+
  # third layer: geom point with nests_clean data
  geom_point(data = nests_clean,
             aes( x = height_cm,
                  y = case_control),
            color = "grey30", # coloring by scientific names
            alpha = 0.4,
            size = 1.5) +
  # separating the panels by type of ant
  facet_wrap (~group) +
  # adjusting color palettes for each panel
  scale_color_brewer (palette = "Set1") +
  scale_fill_brewer(palette = "Set2") +
  # adding labels for x- and y- axes
  labs(
    x = "Tree Height (cm)",
    y = "Predicted Probability of Nest Occurrence") +
  # scale
  scale_y_continuous(labels = scales::percent) +
  # adjusting the theme
  theme_classic() +
  # removing the legend and editing text
  theme(
    legend.position = "none",
    strip.text = element_text(face = "italic", size = 11),
    axis.title = element_text(size = 12))
```



2i.

Figure 1. Species of ant and height of tree influence nesting probability. Separate panels represent different species of ants (left panel is C. sjostedti, middle panel is C. mimosae, and right panel is C. nigriceps). Transparent, grey points represent individual data observations of % probability of nesting (y- axes) from tree height (cm, x-axes) and ant species. Central, opaque line (red, blue, or green [from left to right]) represents interactive logistic model predictions of probability of nesting. Wide, transparent ribbon around line (green, red, blue [left to right]) represents 95% confidence interval for each prediction. Data from: Lujan et al. in 2023. https://doi.org/10.5281/zenodo.8373322

2j.

```
# making new object with tree height of 600 cm from
max_height_data <- data.frame(
  height_cm = c(600, 600, 600),
  scientific_name = factor(c("Crematogaster sjostedti",
                             "Crematogaster mimosae",
                             "Crematogaster nigriceps"),
                           levels = c("Crematogaster sjostedti",
```

```r
                                        "Crematogaster mimosae",
                                        "Crematogaster nigriceps")))
# using ggpredict to get model predictions
predictions_at_600 <- ggpredict(model_intereactive,
                                terms = "scientific_name",
                                condition = c(tree_height_cm = 600))

# displaying table!
print(predictions_at_600)
```

```
# Predicted probabilities of case_control

scientific_name         | Predicted |     95% CI
------------------------------------------------
Crematogaster sjostedti |      0.05 | 0.01, 0.36
Crematogaster mimosae   |      0.19 | 0.14, 0.25
Crematogaster nigriceps |      0.94 | 0.55, 1.00

Adjusted for:
* height_cm = 212.58
```

2k.

Our model demonstrates that tree height and species of ant significantly interact to dictate birds nesting choices. When using the model to predict nest occurrence in the maximum measured height of the trees (600 cm), there was a high chance of it being nested when occupied by Crematogaster nigriceps (0.94 probability, 95% CI[0.55, 1.00]), and low chances if occupied by Crematogaster mimosae (0.19 probability, 95% CI [0.14, 0.25]) or Crematogaster sjostedti (0.05 prbability, 95% CI[0.01, 0.36]). Tree height does play a role, but ant species appears to have a stronger affect on nesting probability seen from the predicted lines. This effect is due to the behavior of the ants. Both Crematogaster nigriceps and Crematogaster mimosae are aggressive, defenssive species, whereas Crematogaster sjostedti tend to be less aggressive. This means that these ants will protect the bird and their young from predators. In addition, C. nigriceps "prune" the trees, which lowers the chance of interactions with other trees.

Problem 3

3a.

In homework 2, I visualized the relationships between cardio vs. workout length and length of workout vs. workout satisfaction, and they're presented on graphs. In homework 3, I visualized the relationship between amount of sleep and satisfaction with workout using circles of varying size and color without a graph.

I noticed a similar color pallet between all my visualizations. I use lots of pink, purples, and blues generally.

I see that my satisfaction with my workout does increase with the more sleep I get, but other factors (workout type, length of workout, etc.) do not impact this vary much. These aren't different between visualizations because the data didn't show different trends.

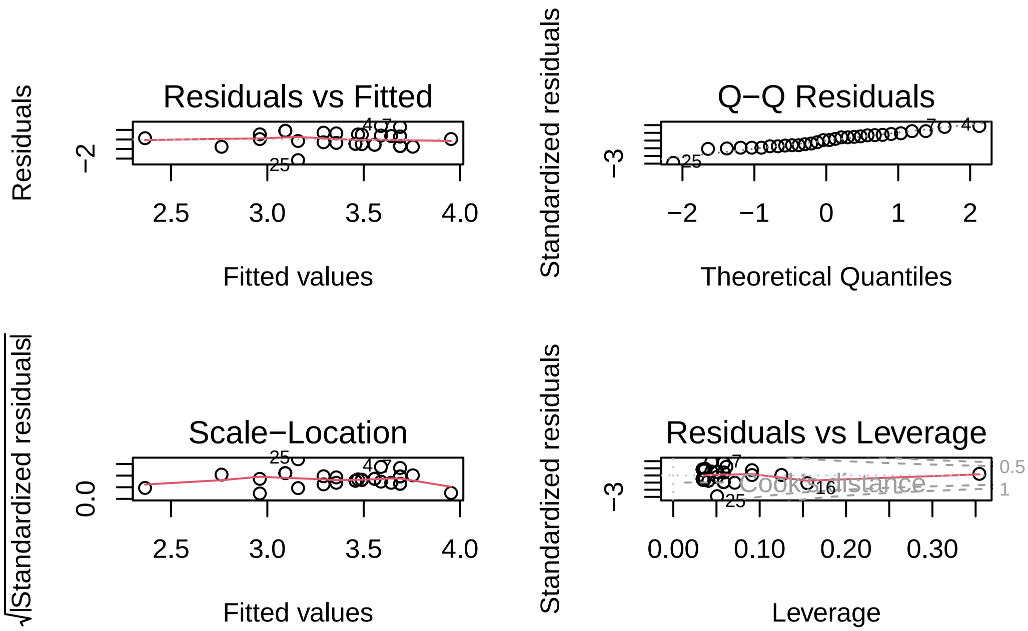
The feedback I got was to not over complicate the affective visualization with too many variables. I implemented this by taking out variables that were superfluous (such as time of meal before workout).

3b.

An appropriate analysis of amount of sleep (continuous numeric variable in hours) and how it effects satisfaction (ordinal variable, measured 1-5) with my workout is a linear regression model. This answers my initial question of "How does the amount of sleep I get effect my satisfaction with my workout?" by showing trends between my workout and the rating I gave the following workout.

3c.

```
# making new object withe data of interest; satisfaction and amount of sleep
# from personal data
workout_model <- lm(satisfaction ~ sleep_hours, data = personal_clean)
# base R residuals
par(mfrow = c(2,2))
plot(workout_model)
```



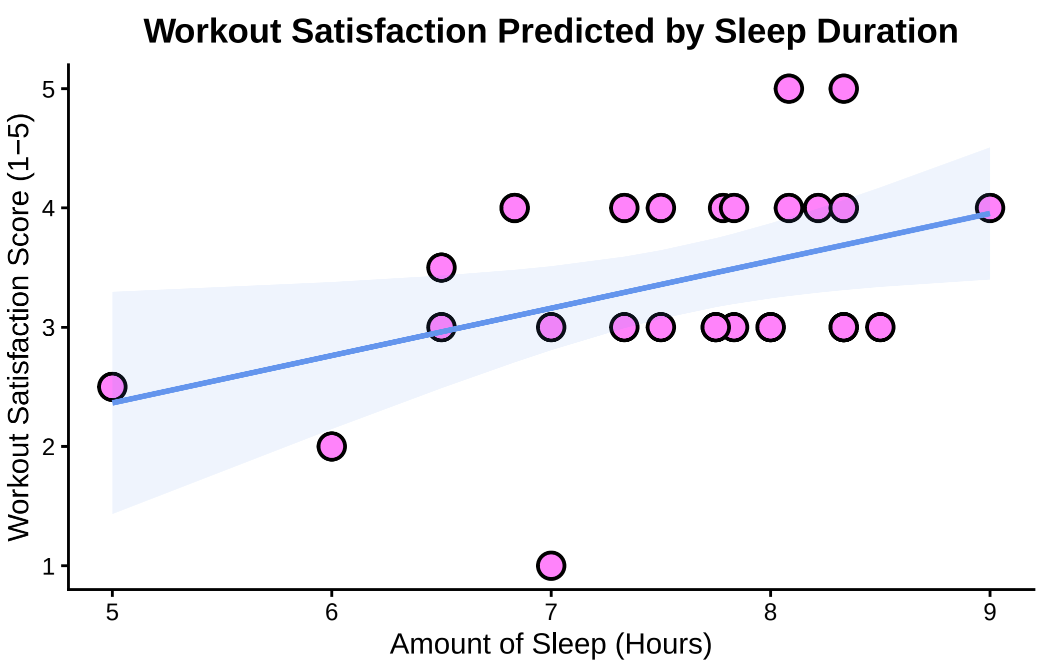

```r
# displaying statistical summary of linear regression
summary(workout_model)
```

```
Call:
lm(formula = satisfaction ~ sleep_hours, data = personal_clean)

Residuals:
     Min       1Q   Median       3Q      Max
-2.15948 -0.54012  0.04263  0.52442  1.41023

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.3791     1.2720   0.298   0.7678
sleep_hours   0.3972     0.1662   2.389   0.0239 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7645 on 28 degrees of freedom
Multiple R-squared:  0.1694,    Adjusted R-squared:  0.1397
F-statistic: 5.709 on 1 and 28 DF,  p-value: 0.02385
```

3d.

```r
# using ggpredict to get model predictions
personal_preds <- ggpredict(workout_model,
                            terms = "sleep_hours")
#base layer: ggplot
ggplot() +
  # second layer: geom_point to show underlying data with personal_clean data
  geom_point( data = personal_clean,
       aes(x = sleep_hours,
       y = satisfaction),
       fill = "orchid1",
       size = 4,
       stroke = 1,
       shape = 21) +
  # third layer: geom_ribbon to show CI
 # using predictions data frame
 geom_ribbon(data = personal_preds,
             aes(x = x,
                 y = predicted,
                 ymin = conf.low,
                 ymax = conf.high),
             fill = "cornflowerblue",
             alpha = 0.1) +
  # third layer: line representing model predictions
  geom_line(data = personal_preds,
            aes(x = x,
                y = predicted),
            color = "cornflowerblue",
            linewidth = 1) +
  # adding axes titles and titles!
  labs(
    x = "Amount of Sleep (Hours)",
    y = "Workout Satisfaction Score (1-5)",
    title = "Workout Satisfaction Predicted by Sleep Duration") +
  # changing theme
  theme_classic() +
  # removing grid lines, bolding title, and changing axes text
  theme(
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
```

```
    plot.title = element_text(face = "bold", size = 13, hjust = 0.5),
    axis.title = element_text(face = "plain", size = 11))
```



3e.

Figure 2. Less sleep is associated with lower workout satisfaction. Data collected from workouts taken place in the time period 4-20-2026 to 5-27-2026 (Data accessed 05-28-2026). Orchid points represent individual observations of amount of sleep (hours) (total n = 30) plotted against workout satisfaction score (1-5). Cornflower blue, thin line represents the linear regression model prediction for workout satisfaction given amount of sleep. Thick, transparent, cornflower blue ribbon represents 95% confidence interval of linear regression model.

3f.

It was found that amount of sleep acted as a statistically significant predictor of my satisfaction with my workout (linear regression, $F(28, 1) = 5.709$, $R^2 = 0.1694$, $p = 0.02385$, $\alpha = 0.05$). Based on the model predictions, for each 1 unit increase in sleep, it's expected to have a increase of 0.3972 ± 0.1662 in workout satisfaction. Approximately 16.94% of total variance in satisfaction scores is accounted for by the moderate effect size sleep duration has ($R^2 = 0.1694$).